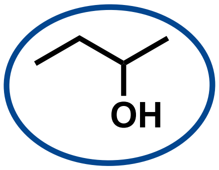
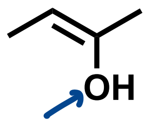
Chem 0345 – Organic Lab
Activity 6: Dehydration of Cyclohexanol
Extra Practice Problems – KEY

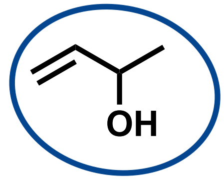
1. For each molecule below, determine if it can undergo E1 to form an alkene when combined with a non-nucleophilic acid. Briefly explain why or why not. If circled, then molecule can undergo E1



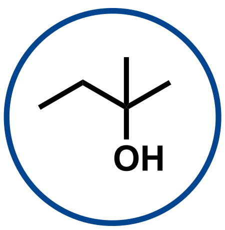
will make 2° C⊕



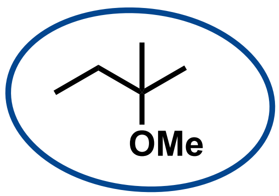
even if protonated to make good leaving group, cannot leave because it is attached to sp^2-hybridized C and will not form stable C⊕



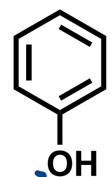
will make resonance-stabilized 2° C⊕



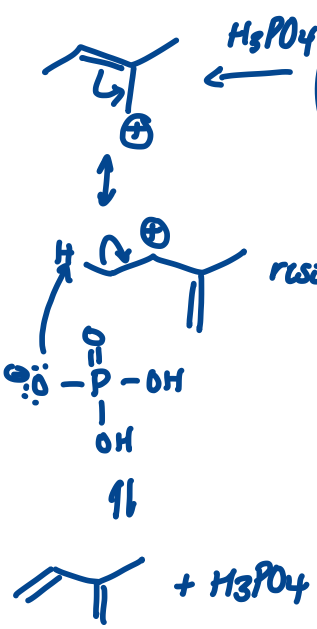
will make 3° C⊕

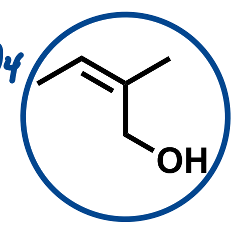


will make 3° C⊕

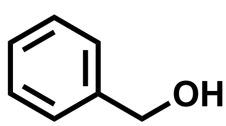


even if protonated to make good leaving group, cannot leave because it is attached to sp^2-hybridized C and will not form stable C⊕

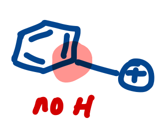


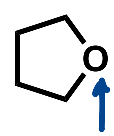


will make resonance-stabilized 1° C⊕



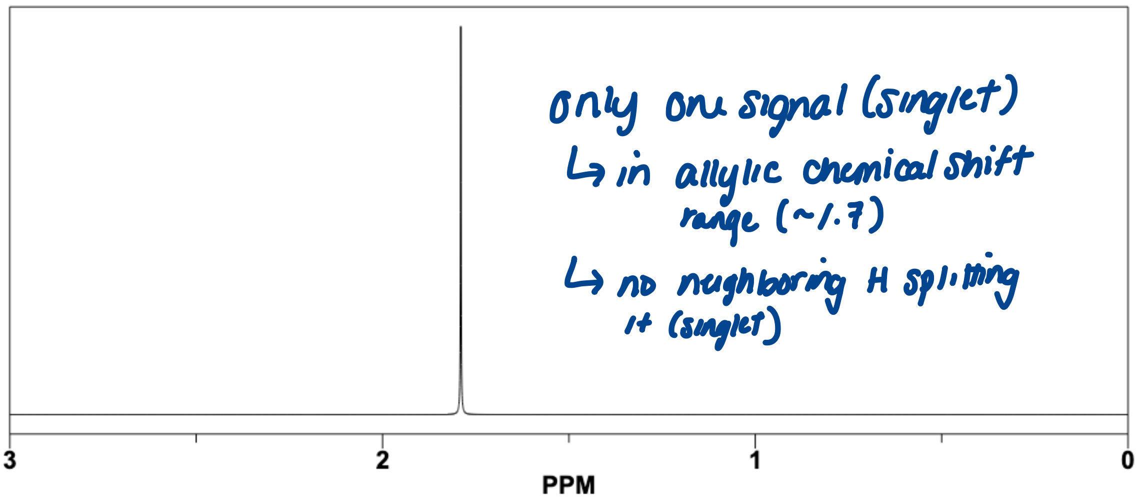
can make resonance-stabilized 1° C⊕, but no H to deprotonate and form alkene:





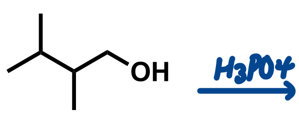
even if protonated to make good leaving group, cannot leave because it is attached to a primary (1°) C and will not form stable C⊕

2. A substrate is combined with phosphoric acid and heated to form a single product. An NMR spectrum was then acquired of the product. Which molecule below was the substrate? Assume that the only signals that were observed for this molecule are in the spectrum are shown.



a) 

1 signal = 12 H, singlet

b) 

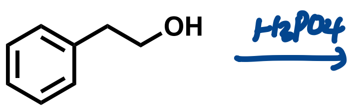
no rxn (1° C⊕ if leaving group leaves)

c) 

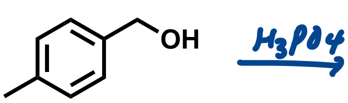
4 signals

d) 

6 signals
(aromatic, vinyl, and allylic)

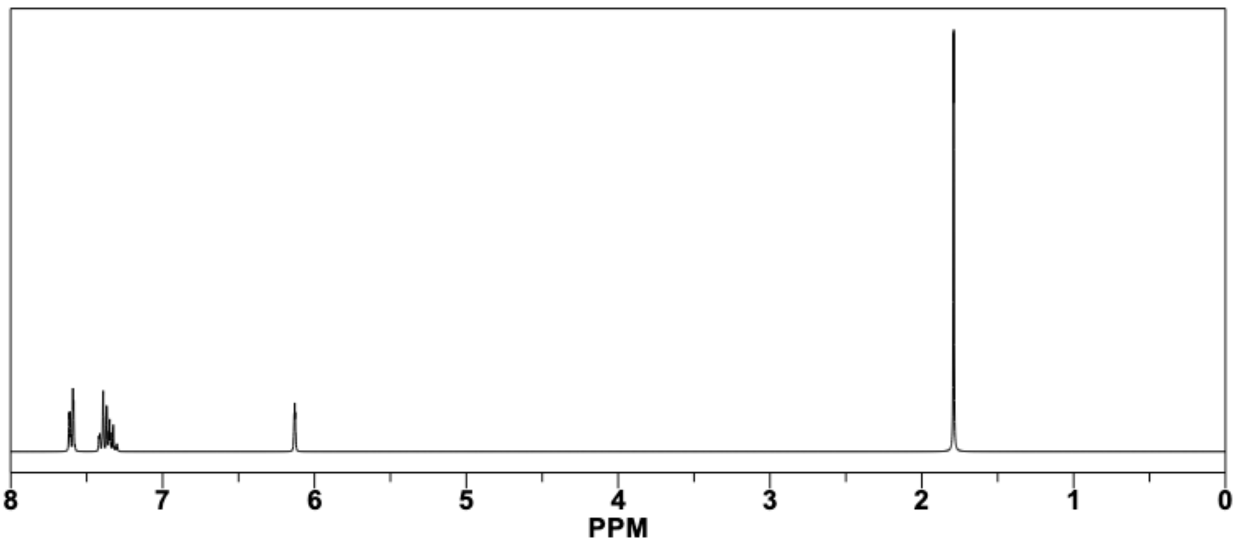
e) 

no rxn (1° C⊕ if leaving group leaves)

f) 

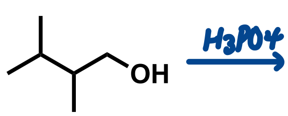
no rxn (no H adjacent to C⊕ to eliminate; 1° C⊕ if leaving group leaves)

3. A substrate is combined with phosphoric acid and heated to form a single product. An NMR spectrum was then acquired of the product. Which molecule below was the substrate? Assume that the only signals that were observed for this molecule are in the spectrum are shown.



a) 

1 signal = 12 H, singlet

b) 

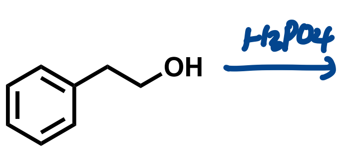
no rxn (1° C⊕ if leaving group leaves)

c) 

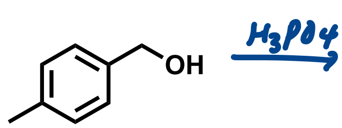
4 signals

d) 

6 signals
(aromatic, vinyl, and allylic)

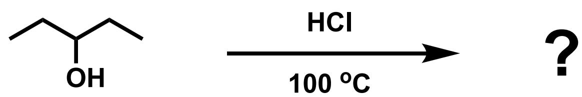
e) 

no rxn (1° C⊕ if leaving group leaves)

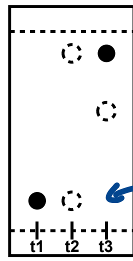
f) 

no rxn (no H adjacent to C⊕ to eliminate; 1° C⊕ if leaving group leaves)

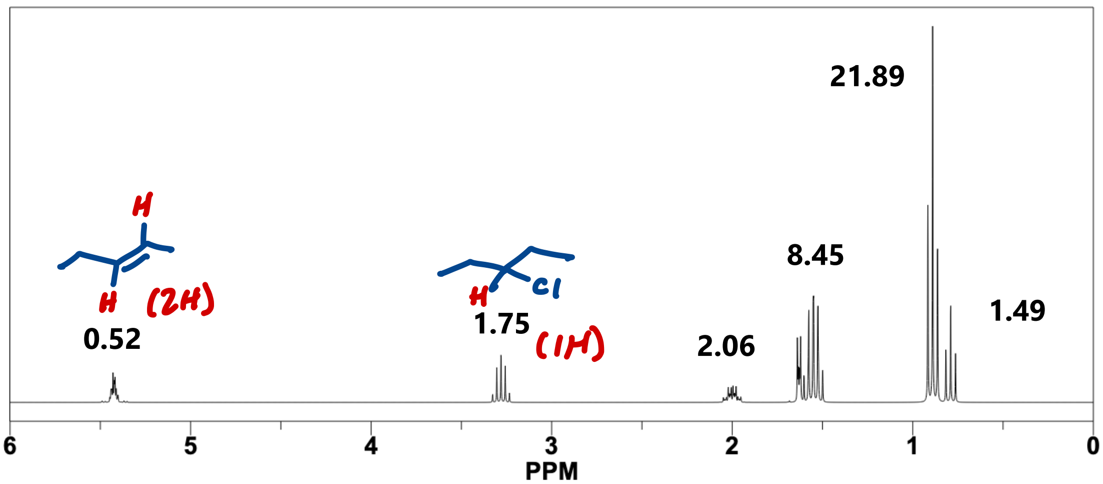
4. Consider the reaction below:



The following TLC was developed at various time points during the reaction (t1, t2, t3) and then an NMR spectrum was taken of the final reaction mixture:



starting material all consumed



a) At what time (t1, t2, or t3) is all the 3-pentanol consumed?

t3

b) What product(s) are formed during the reaction?

c) What product is formed first?

(higher Rf spot appears at t2)
↳ less polar molecule

d) Assuming that only two products are formed, what percentage of the final mixture is the major product?

$$\% \text{ alkene} = \frac{\left(\frac{0.52}{2H}\right)}{\left(\frac{0.52}{2H}\right) + \left(\frac{1.75}{1H}\right)}(100) = 12.9\%$$

(major product)
$\%$ 3-chloropentane =
$100\% - 12.9\%$
$= 87.1\%$